

Cross-Country PII Exposure and Risk Analysis

A synthesis of storage-level privacy findings for thirteen country personas

PII Statistical Modeling – Country Extension

October 2, 2026

Abstract

This document synthesizes the results of the country-level extension of the PII statistical modeling pipeline, covering thirteen synthetic personas (IT_0573, LU_0634, PT_0838, SE_0964, ES_0290, DE_0018, FR_0429, IE_0504, NL_0694, PL_0742, AT_0077, DK_0199, FI_0373) browsing under two authentication states (AUTH / NOTAUTH) with a single consent policy (PARTIAL, labeled *Essential Only* in the figures below). It covers data volumes, PII category prevalence, storage-type distribution, cookie security non-conformity, and the exposure–impact–risk ($P_i \times I_i = R_i$) scoring model, and closes with an explicit discussion of the methodology’s accuracy and limitations. The country pipeline is fully separate from the original three-user French (FR) pipeline; no FR script, data, or published result was modified in the course of this work. This revision extends an earlier six-country version of the same report with seven additional personas (FR_0429, IE_0504, NL_0694, PL_0742, AT_0077, DK_0199, FI_0373); every number below is recomputed from the full thirteen-country dataset, not merged with the earlier figures.

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1 Introduction and Scope

The goal of this analysis is to compare how much personal data, of what kind, and under what technical protection, is collected by websites across thirteen country contexts, and whether authentication status changes that exposure. Each persona is a synthetic identity (*Chris Martin*) carrying country-specific contact details, browsing a set of country-appropriate sites while their browser’s cookies, `localStorage`, `sessionStorage`, and `IndexedDB` stores were captured.

Table 1: Scope of the country dataset.

Dimension	Value
Countries (personas)	IT_0573, LU_0634, PT_0838, SE_0964, ES_0290, DE_0018, FR_0429, IE_0504, NL_0694, PL_0742, AT_0077, DK_0199, FI_0373
Authentication states	AUTH, NOTAUTH
Consent policy	PARTIAL / “Essential Only” (only)
Storage types	cookies, localStorage, sessionStorage, IndexedDB
Lifecycle events tracked	added, modified, removed
Profiles analyzed	26 (13 countries $\times$ 2 auth states)

Unlike the original French dataset – which was collected under three consent policies (ALL/PARTIAL/NONE) – the country dataset only has **PARTIAL**. Consequently, every comparison that in the FR pipeline ran across consent policies has no equivalent here; this document instead adds a comparison axis the FR pipeline never had: side-by-side comparison across the thirteen countries themselves.

2 Methodology Overview

The pipeline runs in three stages, each fully separated from the FR pipeline by directory (no FR script or output is read or written by any stage below).

Stage 1. Categorization. Raw captures are labeled into PII categories (e.g. `DIRECT_PII`, `BEHAVIORAL_DATA`, `IDENTITY_TRACKING`) first by regex matching against the personas’ known ground-truth identity fields and generic tracking-keyword patterns, then by an LLM (`gpt-4.1-mini`) pass over whatever regex left `UNCATEGORIZED`, followed by a dedicated correction pass and a final hand-audited regex pass for third-party SDK field-naming schemes.

Stage 2. Descriptive analysis. Category prevalence, storage-type breakdown, cookie lifetime, cookie security attributes (`httpOnly/secure/sameSite`), and third-party provenance are computed per profile and merged into one cross-country comparison.

Stage 3. Risk scoring. Each physical item is scored on a technical exposure probability P_i and a harm/impact probability I_i , combined multiplicatively into a risk score $R_i = P_i \times I_i \in [0, 1]$.

The thirteen personas were onboarded in three batches as data collection progressed (six, then one, then six more); the categorization pipeline is designed so that onboarding a new batch only ever processes that batch’s own data (via a `-users` scope on every stage script) and never re-runs or overwrites the already-finalized categorization of previously onboarded personas. The descriptive-analysis and risk-scoring stages, by contrast, are pure derived computation and are re-run over the full roster every time a batch is added, which is why every number in this document is already computed over all thirteen personas together.

2.1 Exposure – Impact – Risk model

The exposure model is a logistic function of technical container flags:

$$P_i = \sigma(\eta_i), \quad \eta_i = \beta_0 + \beta_{ho}x_{ho} + \beta_{se}x_{se} + \beta_{ss}x_{ss} + \beta_{tp}x_{tp} + \beta_{pe}x_{pe} + \beta_{ho-pe}(x_{ho}x_{pe}) \quad (1)$$

where x_{ho} = JS-accessible (not `httpOnly`), x_{se} = network-exposed (not `secure`), x_{ss} = cross-site (`sameSite=None`), x_{tp} = third-party, and x_{pe} = persistent (≥ 6 months). The coefficients were calibrated once, offline, via ridge regression in logit space against 13 hand-scored expert scenarios, and are reused unchanged for the country dataset (Table 2).

Table 2: Calibrated exposure coefficients (shared with the FR pipeline, not re-fit).

Coefficient	Value
Intercept (β_0)	−2.257
β_{ho} (JS-accessible)	2.1291
β_{se} (network-exposed)	1.2413
β_{ss} (cross-site)	1.0986
β_{tp} (third-party)	1.2208
β_{pe} (persistent)	0.8172
β_{ho-pe} (interaction)	0.0365

The impact model combines, via a noisy-OR, a 6-dimensional harm vector $z = [z_{ID}, z_{ATO}, z_{LINK}, z_{LOC}, z_{PROF}, z_{ENV}]$ (identification, account takeover, linkability, location, profiling, device fingerprint) that is hand-mapped per PII category, weighted by fixed dimension-level detection probabilities α :

$$I_i = 1 - \prod_k (1 - \alpha_k z_k), \quad \alpha = \{0.90, 0.95, 0.85, 0.75, 0.70, 0.50\} \quad (2)$$

Sensitivity analysis on these weights was explicitly excluded from this phase of work – the figures below should be read as a consistent relative ranking under one fixed set of expert judgments, not as validated absolute probabilities. Section 8 details why.

3 Data Volume and Authentication Impact

Across the 26 profiles, **749 797** items were categorized and a further **446 099** remained `UNCATEGORIZED` after the regex and AI passes (relocated to `data/user_countries_raws/`, so they are excluded from every category-level statistic in this document unless stated otherwise) – a combined total of **1 195 896** observed items.

Table 3: Data volume per country: categorized items, uncategorized items, and the AUTH/NOTAUTH split.

Country	Categorized	Uncategorized	Total (incl. uncat.)	AUTH	NOTAUTH	Ratio N/A
IT_0573	124 620	37 692	162 312	50 831	75 636	1.49
LU_0634	59 360	83 107	142 467	29 376	30 615	1.04
PT_0838	61 216	104 472	165 688	28 512	33 573	1.18
SE_0964	42 984	11 489	54 473	17 482	25 816	1.48
ES_0290	81 760	21 868	103 628	38 653	44 073	1.14
DE_0018	35 304	23 010	58 314	11 265	24 364	2.16
FR_0429	99 708	47 276	146 984	21 362	81 034	3.79
IE_0504	38 738	5 127	43 865	22 218	17 145	0.77
NL_0694	19 815	1 748	21 563	8 796	11 427	1.30
PL_0742	33 623	24 149	57 772	12 143	22 136	1.82

Country	Categorized	Uncategorized	Total (incl. uncat.)	AUTH	NOTAUTH	Ratio N/A
AT_0077	86 806	33 471	120 277	49 734	38 586	0.78
DK_0199	31 454	21 524	52 978	14 583	17 087	1.17
FI_0373	34 409	31 166	65 575	25 429	9 208	0.36
Total	749 797	446 099	1 195 896	330 384	430 700	1.30

Ratio N/A is the NOTAUTH/AUTH ratio of categorized items. Pooled, NOTAUTH items (430 700) outnumber AUTH items (330 384) by a ratio of **1.30**; including **UNCATEGORIZED**, the ratio widens slightly to **1.36** (AUTH: 506 497; NOTAUTH: 689 399). This global ratio, however, hides a reversal that did not appear in the earlier six-country sample: the per-country ratio ranges from **3.79** (FR_0429, strongly NOTAUTH-heavy) down to **0.36** (FI_0373), and for three countries – IE_0504, AT_0077, FI_0373 – *AUTH* browsing actually leaves a larger technical footprint than NOTAUTH. “Non-authenticated browsing collects more data” is therefore not a universal pattern in this dataset; it is the majority pattern (ten of thirteen countries), with three clear exceptions.

Inter-country variability on categorized volume alone spans a factor of **6.29** between the highest (IT_0573, 124 620) and lowest (NL_0694, 19 815) totals – see Section 8 for why this should not be read as a direct measure of “how much more IT_0573 is tracked.”

4 PII Category Prevalence

BEHAVIORAL_DATA is the single largest category overall and remains the top category in 10 of the thirteen countries, but it is not universal: IE_0504 (IDENTITY_TRACKING), DK_0199 (IDENTITY_TRACKING), FI_0373 (ID_SOLUTIONS_AND_EXCHANGES) are instead led by a tracking or ad-exchange category. Table 4 gives the share of each country’s categorized items taken by the eight globally most common categories.

Table 4: Category share (% of each country’s categorized items) for the eight most common categories globally, plus everything else. Columns are two-letter country codes: IT=IT_0573, LU=LU_0634, PT=PT_0838, SE=SE_0964, ES=ES_0290, DE=DE_0018, FR=FR_0429, IE=IE_0504, NL=NL_0694, PL=PL_0742, AT=AT_0077, DK=DK_0199, FI=FI_0373.

Category	IT	LU	PT	SE	ES	DE	FR	IE	NL	PL	AT	DK	FI
BEHAVIORAL_DATA	54.1	37.9	46.5	50.6	39.9	30.8	54.9	21.0	33.0	29.4	38.1	14.5	7.0
IDENTITY_TRACKING	13.9	24.2	19.8	15.8	20.4	21.4	20.6	25.0	19.5	28.2	20.4	25.1	22.5
ID_SOLUTIONS	8.3	4.6	12.5	7.5	10.3	19.7	4.0	13.9	4.0	11.8	5.6	9.2	41.9
DEVICE_ENV	13.1	12.7	6.5	11.5	7.3	5.5	8.7	1.6	5.1	4.3	7.0	2.2	2.9
UX_ANALYTICS	1.8	5.9	3.3	3.5	6.3	5.9	2.3	13.6	17.1	7.8	3.8	17.0	9.9
USER_PREFS	1.7	2.9	2.5	1.4	2.2	5.2	2.4	5.7	3.1	3.3	2.7	6.6	3.0
LOCATION_AND_DEMOGRAPHICS	0.6	0.2	0.4	2.1	0.5	1.3	0.8	1.9	0.7	1.8	13.4	1.5	0.8
SECURITY_BOT	1.2	1.5	1.3	1.6	1.9	1.8	0.6	3.3	1.6	2.6	1.2	13.6	2.4
Other	5.1	10.1	7.1	6.1	11.3	8.4	5.7	14.0	16.0	10.9	7.9	10.3	9.5
N (items)	125 554	59 946	62 014	43 224	82 496	35 508	102 374	39 363	20 223	34 279	88 320	31 670	34 637

DEVICE_ENV (fingerprinting-adjacent signals) is the category with the widest spread: it reaches 13.1% of IT_0573’s items and 12.7% of LU_0634’s, against 1.6% for IE_0504. With thirteen countries, this is no longer a single outlier against a flat baseline – it is a smooth, roughly eight-fold gradient – which is a different and more informative picture than the earlier six-country sample suggested, and still warrants a per-site look before drawing a country-level conclusion from it (see Section 8 on the site-selection confound).

5 Storage-Type Distribution

IndexedDB is the largest single storage type in eleven of the thirteen countries, ranging from 89.6% of FR_0429’s items down to 35.1% for IE_0504 – a much wider range than the earlier six-country sample showed, and no longer a uniform “wide margin everywhere” pattern. Where IndexedDB does

Table 5: Item volume by storage type and country (AUTH + NOTAUTH combined).

Country	Cookies	localStorage	sessionStorage	IndexedDB
IT_0573	8 391	4 342	1 965	111 769
LU_0634	8 126	3 162	822	47 881
PT_0838	9 222	2 802	886	49 175
SE_0964	5 912	1 989	464	34 933
ES_0290	19 492	9 391	2 939	50 904
DE_0018	7 059	3 404	974	24 192
FR_0429	7 453	2 397	820	91 726
IE_0504	11 545	6 375	7 640	13 803
NL_0694	8 072	1 741	1 777	8 633
PL_0742	11 026	3 865	1 648	17 740
AT_0077	10 201	3 751	995	73 373
DK_0199	8 386	3 414	1 226	18 644
FI_0373	9 961	6 098	1 027	17 551

dominate, it is why **BEHAVIORAL_DATA** – its dominant category, largely game- and media-site state caches – leads the global category ranking (Section 5 above). This also means country-level totals are highly sensitive to whether any one persona happened to visit an **IndexedDB**-heavy site (a game portal, a video platform); see Section 8.

6 Cookie Security Non-Conformity

Of the **124 846** cookies analyzed across all thirteen countries, only **23.6%** are fully conformant (**httpOnly** and **secure**); **32.6%** set neither flag. **httpOnly** is set on only **24.6%** of cookies overall, **secure** on **66.5%**, and **62.3%** of cookies are third-party.

Table 6: Cookie security attributes by country (% of that country’s cookies).

Country	<i>n</i> cookies	httpOnly	secure	3rd-party	Secure	Partial	Lowly Secure
IT_0573	8 391	24.8%	61.4%	51.0%	23.2%	39.6%	37.1%
LU_0634	8 126	27.3%	66.2%	67.5%	25.7%	42.1%	32.2%
PT_0838	9 222	26.5%	63.0%	61.7%	24.7%	40.1%	35.2%
SE_0964	5 912	32.2%	67.3%	70.0%	31.2%	37.1%	31.7%
ES_0290	19 492	25.5%	73.7%	66.1%	25.1%	49.0%	25.9%
DE_0018	7 059	20.2%	67.2%	59.7%	19.3%	48.9%	31.8%
FR_0429	7 453	29.7%	69.0%	67.3%	28.4%	41.9%	29.7%
IE_0504	11 545	18.6%	49.6%	43.0%	18.3%	31.6%	50.1%
NL_0694	8 072	19.6%	72.0%	62.6%	18.7%	54.2%	27.1%
PL_0742	11 026	17.2%	57.1%	53.0%	16.4%	41.5%	42.1%
AT_0077	10 201	26.8%	71.3%	72.0%	25.4%	47.4%	27.2%
DK_0199	8 386	30.1%	72.9%	69.0%	29.1%	44.7%	26.1%
FI_0373	9 961	26.0%	72.1%	70.4%	25.2%	47.6%	27.2%
All	124 846	24.6%	66.5%	62.3%	23.6%	43.8%	32.6%

Secure/**Partial**/**Lowly Secure** columns are the derived posture (both flags / one flag / neither flag). **PL_0742** has both the lowest **httpOnly** rate (17.2%) and the lowest fully-secure rate (16.4%) of the thirteen; **SE_0964** has the best **httpOnly** rate (32.2%) and also the highest fully-secure rate (31.2%), while **AT_0077** has the highest third-party exposure (72.0%). No country is uniformly best or worst across every attribute examined – conformity trades off differently in each case, which argues against collapsing this into a single per-country “security score.”

sameSite distribution (not tabulated above for space) is dominated by **No Restriction** or **Lax** in every country; **Strict** ranges from 1.2% (**LU_0634**) to 16.3% (**NL_0694**) of a country’s cookies

– cross-site cookie restriction remains a minority practice everywhere in this dataset, though less uniformly negligible than in the six-country sample.

7 Risk Scoring

7.1 Cookie risk (the only storage type with genuine per-item variation)

Table 7: Cookie risk score (R_i) statistics by country and authentication mode.

Country	Mode	n	Mean	Median	Q1	Q3
IT_0573	AUTH	3 754	0.5691	0.6378	0.4040	0.7918
	NOTAUTH	3 411	0.5195	0.5565	0.2911	0.7918
LU_0634	AUTH	4 304	0.6184	0.6794	0.4557	0.8118
	NOTAUTH	2 524	0.5746	0.6259	0.4021	0.7918
PT_0838	AUTH	4 082	0.6332	0.6794	0.4653	0.8118
	NOTAUTH	3 610	0.5979	0.6794	0.4258	0.7918
SE_0964	AUTH	3 237	0.5826	0.6378	0.4142	0.8118
	NOTAUTH	1 787	0.5504	0.5717	0.3886	0.7918
ES_0290	AUTH	11 160	0.6291	0.6794	0.4653	0.8118
	NOTAUTH	5 396	0.5818	0.6378	0.4198	0.7918
DE_0018	AUTH	2 303	0.6139	0.6794	0.4346	0.8118
	NOTAUTH	3 690	0.5753	0.6378	0.4040	0.7918
FR_0429	AUTH	1 836	0.5910	0.6464	0.4194	0.8118
	NOTAUTH	4 297	0.6180	0.6794	0.4653	0.8118
IE_0504	AUTH	4 425	0.5503	0.6378	0.3467	0.7918
	NOTAUTH	5 110	0.5817	0.6794	0.4068	0.7918
NL_0694	AUTH	2 815	0.5306	0.5047	0.3449	0.7387
	NOTAUTH	3 042	0.5666	0.6221	0.4142	0.7918
PL_0742	AUTH	3 507	0.6351	0.6794	0.4653	0.8118
	NOTAUTH	5 733	0.6370	0.6794	0.4653	0.8118
AT_0077	AUTH	3 530	0.6133	0.6759	0.4396	0.8118
	NOTAUTH	5 075	0.6441	0.7218	0.4653	0.8118
DK_0199	AUTH	3 657	0.5387	0.5409	0.3254	0.7918
	NOTAUTH	3 627	0.5749	0.6378	0.4040	0.8118
FI_0373	AUTH	3 902	0.5680	0.6082	0.3750	0.8118
	NOTAUTH	4 703	0.5765	0.6082	0.4040	0.8118

Pooled across all storage types, mean R_i is 0.5894 under AUTH and 0.5945 under NOTAUTH – the two modes remain close, as in the earlier sample, even though the countries now span a much wider range of *volumes* (Section 5). For cookies specifically (Table 7), AUTH scores a slightly higher or comparable median risk to NOTAUTH in most countries – opposite of the volume pattern, where NOTAUTH usually (but not always, Section 5 above) has more items – authenticated sessions tend to set fewer cookies overall, but a comparable or larger share of the ones they do set carry higher-impact categories or weaker technical protections.

7.2 Risk score by storage type – a structural limitation

Table 8’s `IndexedDB` column is identical or near-identical across most countries. This is not a finding about the data – it is a direct consequence of the exposure model treating `IndexedDB` exposure flags as fixed constants rather than measured per item (Section 8). The apparent uniformity of `IndexedDB` risk across countries should not be read as “`IndexedDB` exposure is genuinely identical everywhere.”

7.3 Figures

Per-country versions of Figure 2 (26 figures, one per country/mode combination) are provided in Appendix A.

Table 8: Median risk score (R_i) by storage type and country, AUTH mode.

Country	Cookie	localStorage	sessionStorage	IndexedDB
IT_0573	0.6378	0.5577	0.3874	0.5577
LU_0634	0.6794	0.6082	0.3874	0.5577
PT_0838	0.6794	0.6082	0.3874	0.5577
SE_0964	0.6378	0.5577	0.4397	0.5577
ES_0290	0.6794	0.6545	0.3874	0.5577
DE_0018	0.6794	0.6436	0.3874	0.5577
FR_0429	0.6464	0.5577	0.3874	0.5577
IE_0504	0.6378	0.5577	0.4470	0.5577
NL_0694	0.5047	0.5577	0.3874	0.5577
PL_0742	0.6794	0.5577	0.3874	0.5577
AT_0077	0.6759	0.6082	0.3874	0.5577
DK_0199	0.5409	0.6082	0.3874	0.5577
FI_0373	0.6082	0.6082	0.3874	0.7898

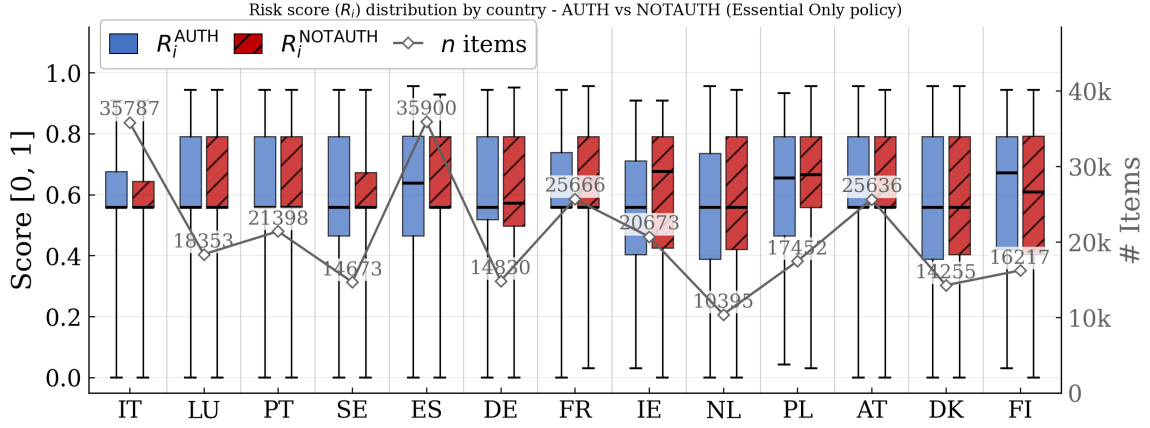


Figure 1: Risk score (R_i) distribution across the thirteen countries, AUTH vs NOTAUTH, all storage types pooled, Essential Only (PARTIAL) policy. Line overlay is item count per country.

8 Discussion and Limitations

This section states plainly what can and cannot be concluded from the numbers above.

8.1 What is solid

The literal DIRECT_PII matches (email, phone, name, address, date of birth) are matched against exact strings known from each synthetic persona – there is no ambiguity and no false-positive risk for these specific fields. This is the one part of the pipeline with genuine ground truth.

8.2 Categorization-layer caveats

- A regex bug that matched long digit chains (e.g. TCF consent strings) as IP addresses was found and fixed for the country pipeline only; whether the original FR pipeline carries the same bug was out of scope and was not checked.
- A duplication bug in cookie categorization caused some subcategories (e.g. **gender**) to be counted up to $\sim 29\times$ per real occurrence; fixed for the country pipeline. Its status in the FR pipeline is likewise unverified.

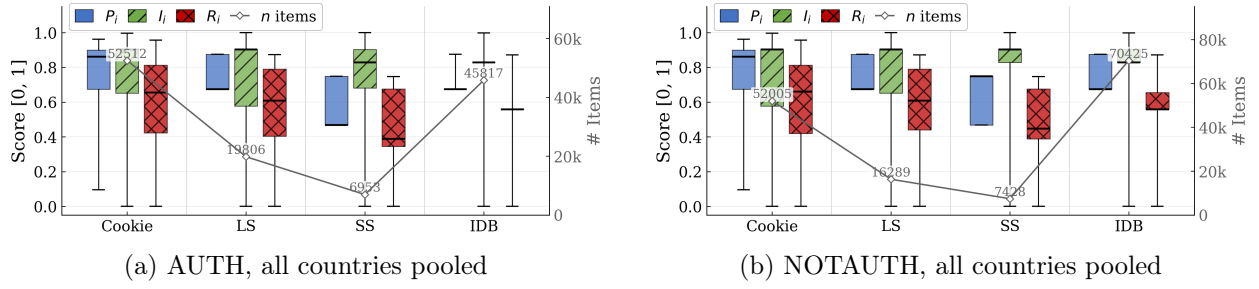


Figure 2: Exposure (P_i), impact (I_i), and risk (R_i) distributions by storage type, pooled across all thirteen countries.

- DIRECT_PII_KEYS’s `user_id_key` pattern is known to be dominated by non-PII matches (consent-platform tokens, generic numeric session IDs, the literal string "anonymous") – this was identified but left unfixed by explicit decision.
- The first LLM categorization pass batches IndexedDB records by record count rather than field count, so records with 50+ uncategorized fields could silently exceed the model’s output limit and lose fields with no error raised. A dedicated field-capped correction pass was added specifically to recover these; because that correction pass has since run (and itself mutated the UNCATEGORIZED file in place), the original truncation rate can no longer be reconstructed precisely from the current data for all thirteen countries – it is a documented mechanism, not a number this revision can still quote.

8.3 Risk-model caveats

- Both the exposure coefficients and the impact map are fixed expert judgments, not values fit to observed outcomes. They provide a consistent *relative* ranking under one set of assumptions, not a validated absolute probability of harm.
- **No sensitivity analysis has been run on these weights for the country dataset** (explicitly deferred). Country rankings by risk should be treated as conditional on this one weight set until that analysis is done.
- For `localStorage`, `sessionStorage`, and `IndexedDB`, the exposure features (`js_accessible`, `network_exposed`, `cross_site`, `persistent`) are hardcoded constants per storage type rather than measured per item – only the third-party flag varies. P_i , and hence R_i , is therefore close to a two-valued step function for these three storage types, which is why Table 8 looks largely uniform across countries. Only cookies carry a genuinely continuous exposure score in the current model.
- The impact engine computes an entropy/structuredness “boost” that is never applied (the multiplication is commented out in the ported source) – highly unique or structured values currently receive no impact premium despite the formula suggesting they should.
- $R_i = P_i \times I_i$ assumes exposure and impact are independent, which is a simplification.

8.4 Cross-country comparison caveats

- Each persona completed *one* crawl session on country-appropriate sites; site selection was not held constant across countries. A volume or category difference between two countries may reflect genuinely different tracking practices, or simply which sites (a game portal vs. a grocery site) that persona happened to visit – IT_0573’s outsized `DEVICE_ENV` and `IndexedDB` volumes, and FR_0429’s extreme NOTAUTH/AUTH ratio, are plausible instances of this.

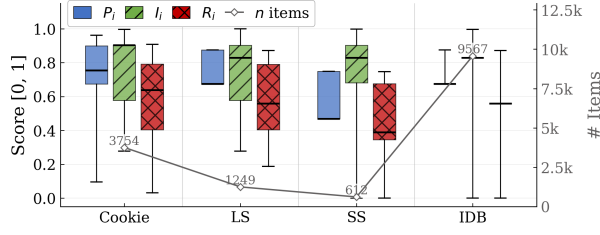
- With one crawl per persona per authentication mode, there is no way to distinguish a stable population-level difference from that single run’s noise. No confidence intervals or significance tests are computed anywhere in this pipeline – every figure above is a descriptive statistic, not an inferential one. With thirteen personas instead of six, the spread of per-country ratios and shares is wider than it first appeared, not narrower – a reminder that adding more single-session personas increases the *range* of outcomes observed without adding any statistical power to generalize from it.
- Third-party detection is a simple domain-suffix check and does not catch more advanced first-party-disguise techniques (e.g. CNAME cloaking); the vendor name list is a static, incomplete hand-authored mapping, so an unrecognized “top vendor” (e.g. `a-mo.net` for ES_0290) may just mean *unclassified domain*, not *identified major tracker*.
- Cookie lifetime/duration figures computed via the lifecycle analysis script are not exactly reproducible across reruns, because that script measures duration against the time the script is run rather than the time of data collection – a pre-existing quirk inherited from the FR codebase. This report does not use that script’s output.

9 Conclusion

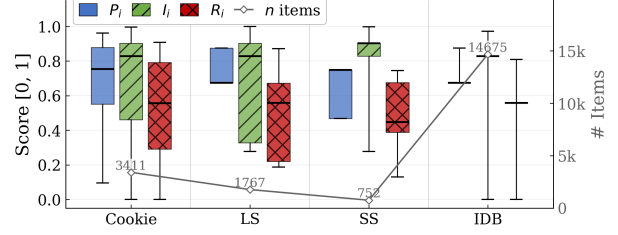
Across the thirteen country personas, tracking-oriented `BEHAVIORAL_DATA` remains the single largest category overall and leads in ten of the thirteen countries, but `IDENTITY_TRACKING` leads in IE_0504 and DK_0199, and `ID_SOLUTIONS_AND_EXCHANGES` leads in FI_0373 – the “`BEHAVIORAL_DATA` everywhere” pattern visible in the earlier six-country sample does not hold once more countries are added. `IndexedDB` is the largest storage type in most, but not all, countries (35.1%–89.6% of items). Authentication status still shifts volume on average – `NOTAUTH` collects $1.30\times$ as many categorized items as `AUTH`, pooled – but that pattern now reverses for three of the thirteen countries (IE_0504, AT_0077, FI_0373), which the six-country sample had no opportunity to reveal. Cookie security is weak everywhere: only about a quarter of cookies in any country are fully conformant (`httpOnly` and `secure` together), and no country is uniformly best or worst across the security attributes examined.

At the risk-score level, per-item severity remains broadly comparable across countries – Figure 1 shows overlapping distributions and similar means (0.5894 `AUTH` vs. 0.5945 `NOTAUTH`, pooled) – so the country-to-country differences visible in the raw data are driven far more by *volume* (how much was collected, which now varies by a factor of $6.29\times$) than by *per-item severity* (how bad each item is). This conclusion is now observed over more than twice as many personas as the original six-country report and still holds, which is some evidence it is not an artifact of the first six personas chosen – but it is also a direct consequence of the current exposure model’s structural limitation for non-cookie storage, and should be revisited once a richer per-item exposure model and a formal sensitivity analysis are in place. Both are natural next steps for this line of work, as is adding a second crawl per persona to separate genuine country-level signal from single-session noise – a gap this revision’s wider, more erratic ratios and shares has made more visible, not less.

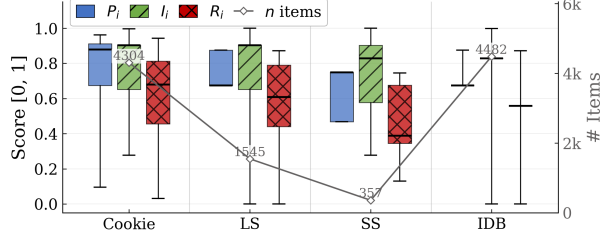
A Per-Country Risk Distribution Figures



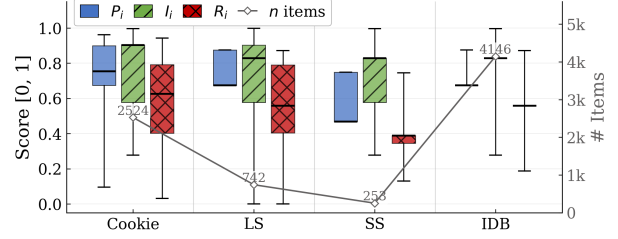
(a) IT_0573 – AUTH



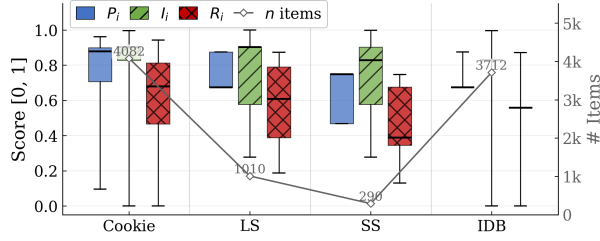
(b) IT_0573 – NOTAUTH



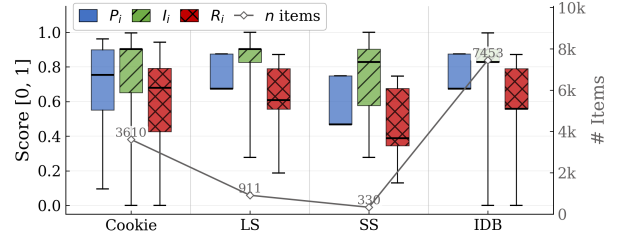
(a) LU_0634 – AUTH



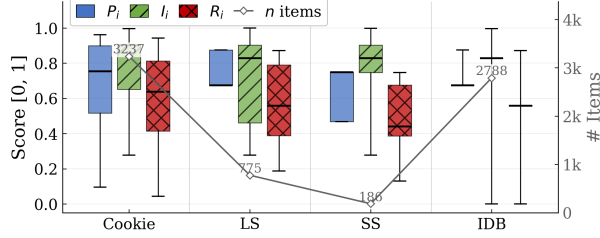
(b) LU_0634 – NOTAUTH



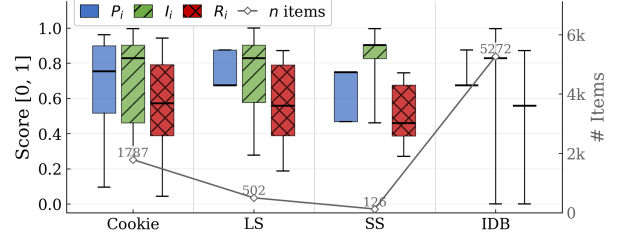
(a) PT_0838 – AUTH



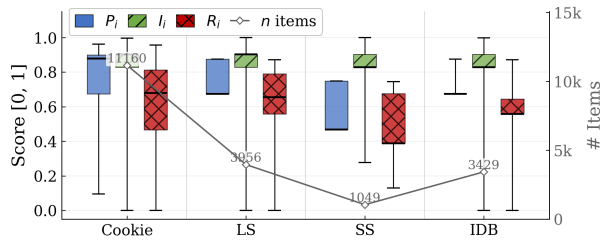
(b) PT_0838 – NOTAUTH



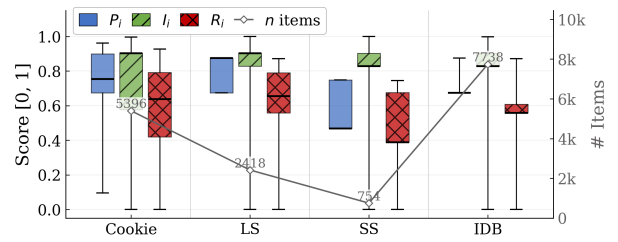
(a) SE_0964 – AUTH



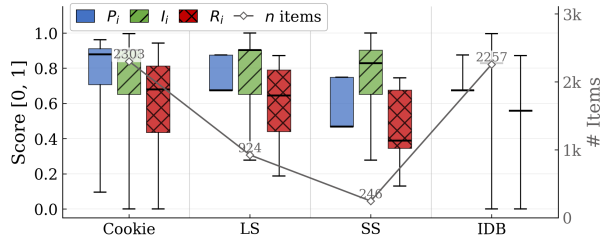
(b) SE_0964 – NOTAUTH



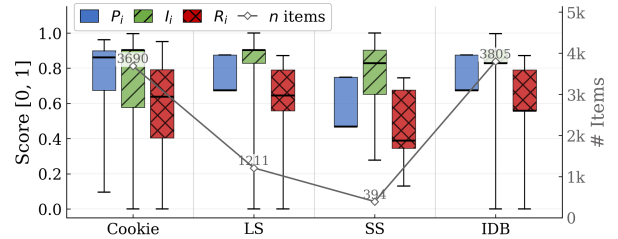
(a) ES_0290 – AUTH



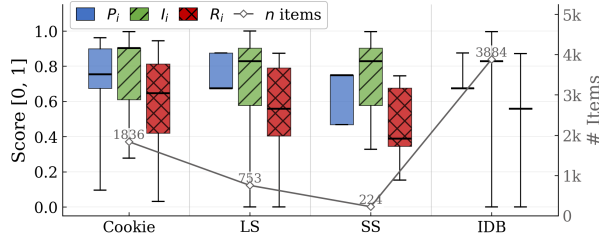
(b) ES_0290 – NOTAUTH



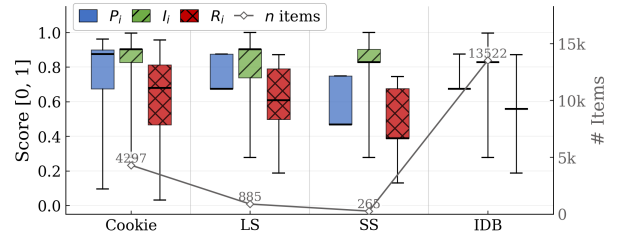
(a) DE_0018 – AUTH



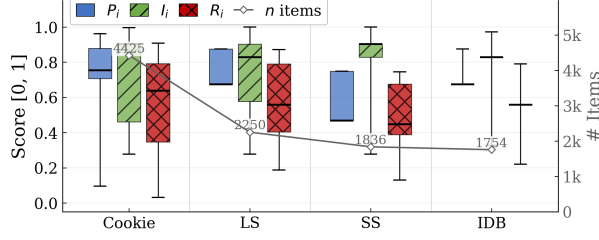
(b) DE_0018 – NOTAUTH



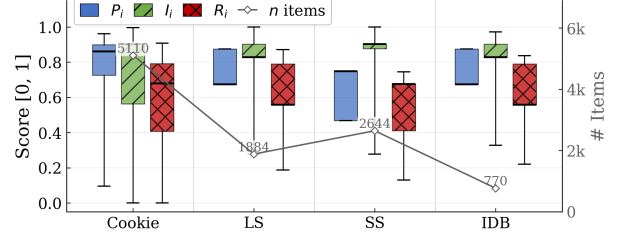
(a) FR_0429 – AUTH



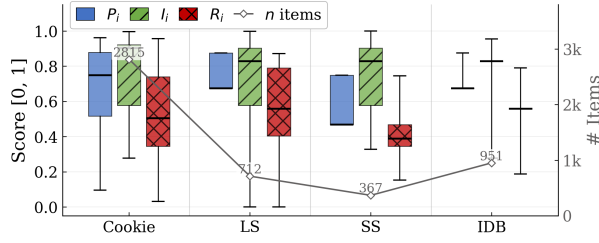
(b) FR_0429 – NOTAUTH



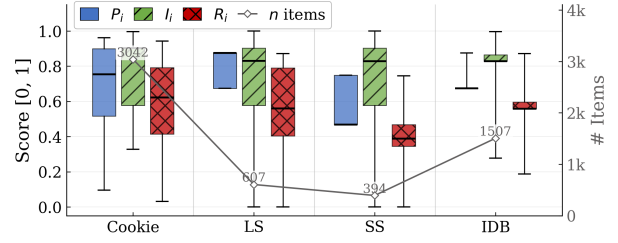
(a) IE_0504 – AUTH



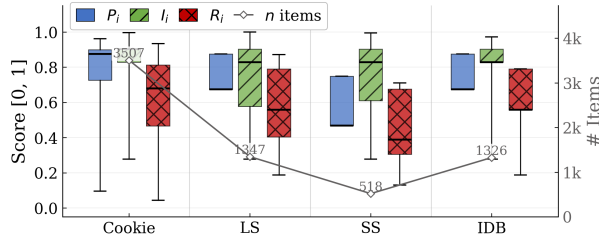
(b) IE_0504 – NOTAUTH



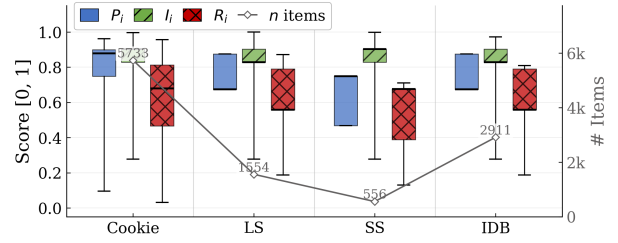
(a) NL_0694 – AUTH



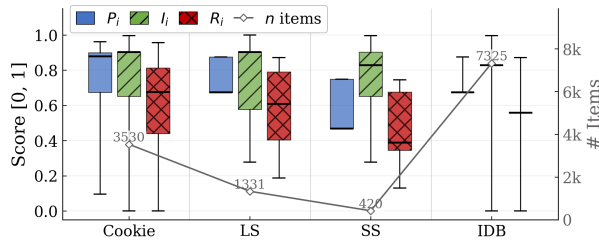
(b) NL_0694 – NOTAUTH



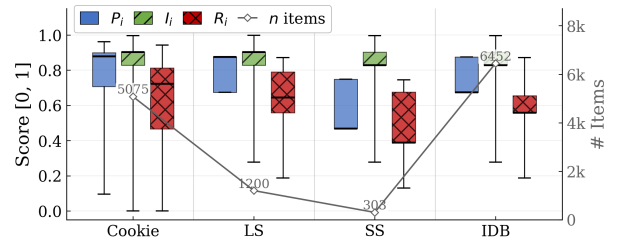
(a) PL_0742 – AUTH



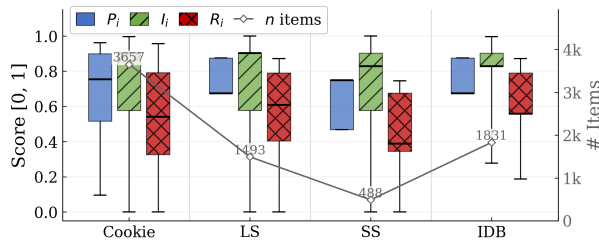
(b) PL_0742 – NOTAUTH



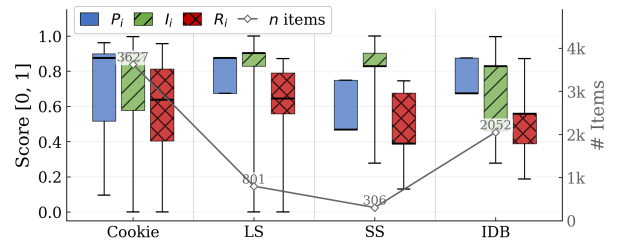
(a) AT_0077 – AUTH



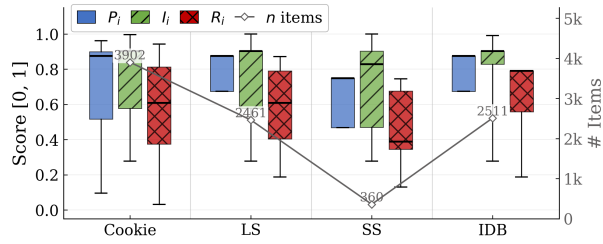
(b) AT_0077 – NOTAUTH



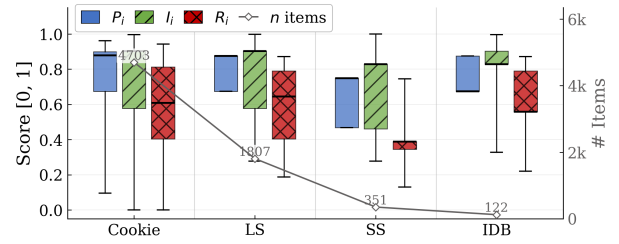
(a) DK_0199 – AUTH



(b) DK_0199 – NOTAUTH



(a) FI_0373 - AUTH



(b) FI_0373 - NOTAUTH